LENA and RtI:
Progressive Progress Monitoring in a Low-Income Neighborhood

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Focus on Language

Language development is fundamental to later academic achievement. The gap between professional families and families receiving welfare is noticeable before age three. Early identification and intervention is crucial to closing this gap, but many communities lack a coordinated system to identify delays and provide effective language interventions for infants and toddlers.

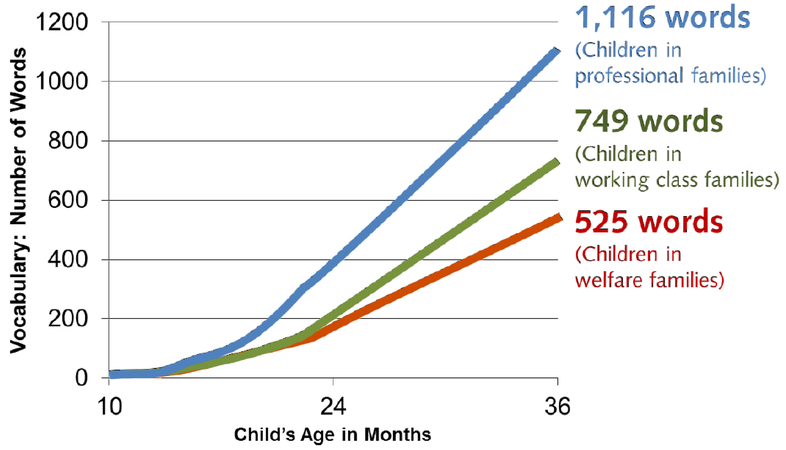
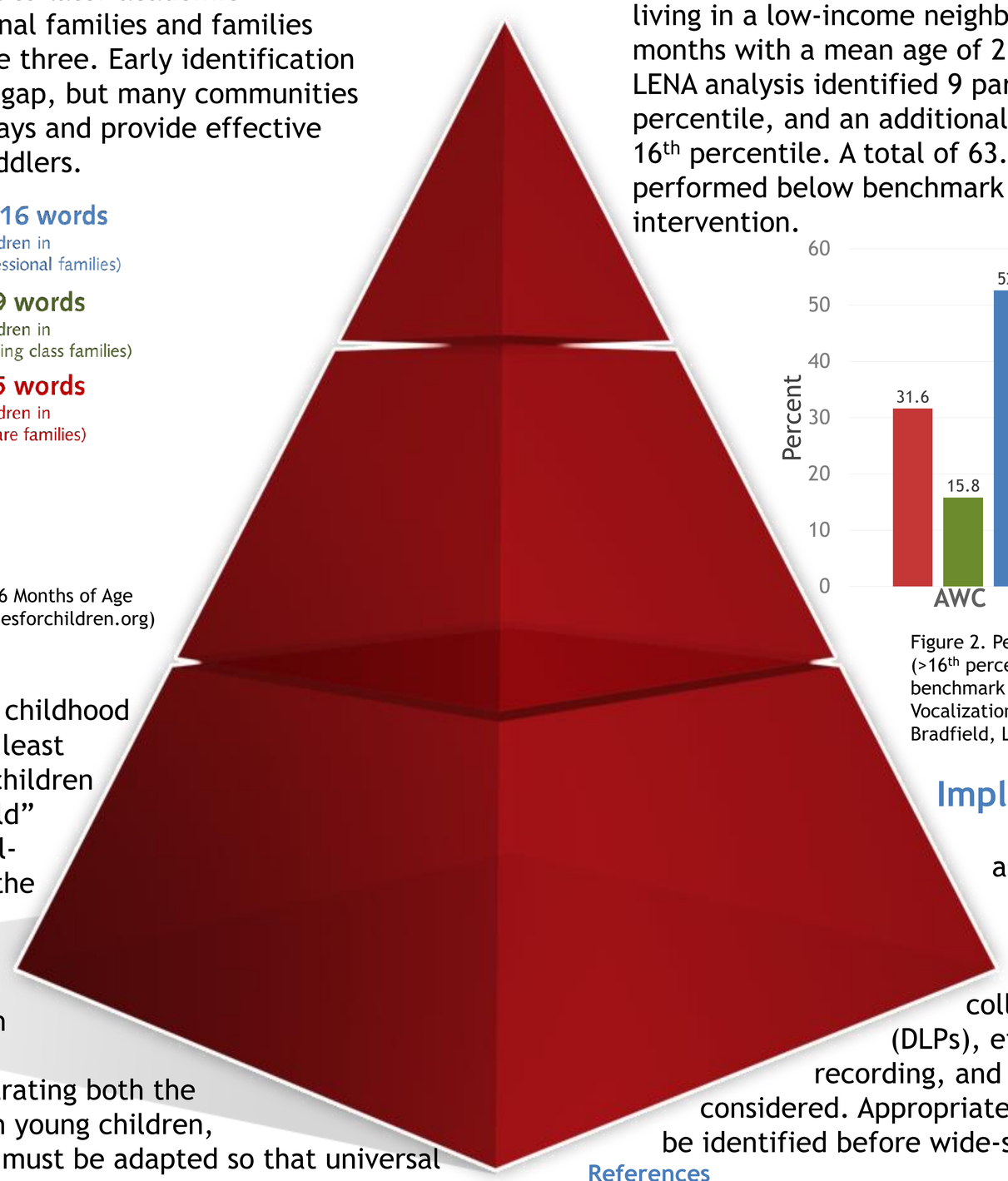


Figure 1. Expressive Vocabulary Estimates in Children 10 to 36 Months of Age (Adapted from Hart & Risley, 1995, available at www.strategiesforchildren.org)

RtI in Community Setting

Minnesota currently mandates, "early childhood developmental screening for children at least once before school entrance, targeting children who are between three and four years old" (Minn. State. § 121A.17, Subd.3(b)). Well-child check-ups are available as part of the Early Periodic Screening, Diagnosis, and Treatment (EPSDT) Program for children who receive Medicaid; unfortunately, these visits are not always effective with families in low-income neighborhoods.

There is a need for research demonstrating both the usefulness and impact of RtI models with young children, particularly for infants and toddlers. RtI must be adapted so that universal screening occurs before age 3, and progress monitoring can be implemented effectively with diverse children and families in a variety early childhood settings within the community. LENA may serve as an effective screening and progress monitoring tool for community-based RtI.



Initial LENA Data

We examined LENA data collected on *n*=19 child-caregiver dyads living in a low-income neighborhood. Child ages ranged from 1 to 47 months with a mean age of 21 months. On one or more measures, LENA analysis identified 9 participants with scores below the 7th percentile, and an additional 3 participants with scores below the 16th percentile. A total of 63.2% of participants in this sample performed below benchmark and may benefit from Tier 2 or 3 intervention.

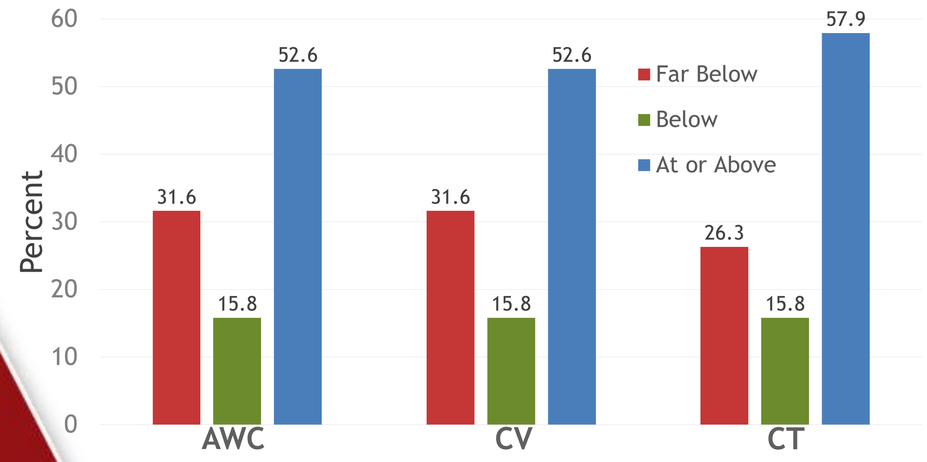


Figure 2. Percent of caregiver-child dyads at or above benchmark (>16th percentile), below benchmark (<16th percentile), or far below benchmark (<7th percentile) on Adult Word Count (AWC), Child Vocalizations (CV), and Child Turns (CT). Data provided by Tracy Bradfield, Lauren Martin, & Alisha Wackerle-Hollman.

Implications for Implementation

There are several challenges associated with the use of LENA as a screening and progress monitoring tool for community-based RtI. The logistics associated with distributing and collecting Digital Language Processors (DLPs), ethical and privacy considerations for recording, and potential sampling error all need to be considered. Appropriate Tier 2 and 3 interventions also need to be identified before wide-scale implementation is possible.

References

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